

Python Programming

Mini Project Assignment

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Branch: AIML

Project 1) Student Management System:

Programming Code:

Student Management System X

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Student Management System > ...

```
1  students = {}
2
3  while True:
4      print("\nStudent Management System:")
5      print("1. Add Student")
6      print("2. Update Marks")
7      print("3. Search Student")
8      print("4. Display All Students")
9      print("5. Remove Student")
10     print("6. Exit")
11
12     choice = int(input("Enter Choice: "))
13
14     if choice == 1:
15         roll = input("Enter Roll No: ")
16         name = input("Enter Name: ")
17         age = int(input("Enter Age: "))
18         marks = float(input("Enter Marks: "))
19
20         students[roll] = {
21             "name": name,
22             "age": age,
23             "marks": marks
24         }
25
```

```
26     elif choice == 2:
27         roll = input("Enter Roll No: ")
28         if roll in students:
29             marks = float(input("Enter New Marks: "))
30             students[roll]["marks"] = marks
31         else:
32             print("Student Not Found")
33
34     elif choice == 3:
35         roll = input("Enter Roll No: ")
36         print(students.get(roll, "Student Not Found"))
37
38     elif choice == 4:
39         for roll, data in students.items():
40             print(roll, ":", data)
41
42     elif choice == 5:
43         roll = input("Enter Roll No: ")
44         students.pop(roll, None)
45         print("Student Removed")
46
47     elif choice == 6:
48         print("Program Ended")
49         break
50
51     else:
52         print("Invalid Choice")
```

Output:

```
OUTPUT    DEBUG CONSOLE    TERMINAL

▼ TERMINAL    Python + ▾ 📄 🗑️ ...

Student Management System:
1. Add Student
2. Update Marks
3. Search Student
4. Display All Students
5. Remove Student
6. Exit
Enter Choice: 1
Enter Roll No: 16
Enter Name: Dhanashri
Enter Age: 19
Enter Marks: 70

Student Management System:
1. Add Student
2. Update Marks
3. Search Student
4. Display All Students
5. Remove Student
6. Exit
Enter Choice: 2
Enter Roll No: 16
Enter New Marks: 75

Student Management System:
1. Add Student
2. Update Marks
3. Search Student
4. Display All Students
5. Remove Student
6. Exit
Enter Choice: 3
Enter Roll No: 16
{'name': 'Dhanashri', 'age': 19, 'marks': 75.0}
```

Student Management System:

1. Add Student
2. Update Marks
3. Search Student
4. Display All Students
5. Remove Student
6. Exit

Enter Choice: 4

16 : {'name': 'Dhanashri', 'age': 19, 'marks': 75.0}

Student Management System:

1. Add Student
2. Update Marks
3. Search Student
4. Display All Students
5. Remove Student
6. Exit

Enter Choice: 5

Enter Roll No: 16

Student Removed

Student Management System:

1. Add Student
2. Update Marks
3. Search Student
4. Display All Students
5. Remove Student
6. Exit

Enter Choice: 6

Program Ended

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Project 2) Library Management System

Programming Code:

```
Library Management System • ▶ ⌵ □ ...
Library Management System > ...

1  library = [
2      [101, "Python Programming"],
3      [102, "Java Programming"],
4      [103, "C Programming"]
5  ]
6
7  while True:
8      print("\nLibrary Management System: ")
9      print("1. Display Books")
10     print("2. Add Book")
11     print("3. Search Book")
12     print("4. Exit")
13
14     choice = int(input("Enter your choice: "))
15
16     if choice == 1:
17         print("\nBook ID\tBook Name")
18         for book in library:
19             print(book[0], "\t", book[1])
20
21     elif choice == 2:
22         book_id = int(input("Enter Book ID: "))
23         book_name = input("Enter Book Name: ")
24         library.append([book_id, book_name])
25         print("Book Added Successfully!")
26
```

```
27 elif choice == 3:
28     search_id = int(input("Enter Book ID to Search: "))
29     found = False
30
31     for book in library:
32         if book[0] == search_id:
33             print("Book Found:", book[1])
34             found = True
35             break
36
37     if not found:
38         print("Book Not Found!")
39
40 elif choice == 4:
41     print("Thank You!")
42     break
43
44 else:
45     print("Invalid Choice!")
```

Output:

```
OUTPUT    DEBUG CONSOLE    TERMINAL

✓ TERMINAL  Python + ▾ □ 🗑 ...

Library Management System:
1. Display Books
2. Add Book
3. Search Book
4. Exit
Enter your choice: 1

Book ID Book Name
101      Python Programming
102      Java Programming
103      C Programming

Library Management System:
1. Display Books
2. Add Book
3. Search Book
4. Exit
Enter your choice: 2
Enter Book ID: 104
Enter Book Name: R Programming
Book Added Successfully!

Library Management System:
1. Display Books
2. Add Book
3. Search Book
4. Exit
Enter your choice: 3
Enter Book ID to Search: 104
Book Found: R Programming
```


Library Management System:

1. Display Books
2. Add Book
3. Search Book
4. Exit

Enter your choice: 3

Enter Book ID to Search: 105

Book Not Found!

Library Management System:

1. Display Books
2. Add Book
3. Search Book
4. Exit

Enter your choice: 4

Thank You!

Project 3) Personal expense tracker

Programming Code:

```
Personal expense tracker ●
Personal expense tracker > ...

1  expenses = []
2  budget = float(input("Enter Monthly Budget: "))
3
4  def add_expense():
5      desc = input("Enter Description: ")
6      category = input("Enter Category: ")
7      amount = float(input("Enter Amount: "))
8      date = input("Enter Date: ")
9
10     expense = {
11         "Description": desc,
12         "Category": category,
13         "Amount": amount,
14         "Date": date
15     }
16
17     expenses.append(expense)
18     print("Expense Added Successfully!")
19
20 def view_expenses():
21     if len(expenses) == 0:
22         print("No Expenses Found.")
23     else:
24         print("\nExpense List")
25         for i, exp in enumerate(expenses, start=1):
26             print(i, exp)
27
```

```
28 def category_summary():
29     summary = {}
30
31     for exp in expenses:
32         cat = exp["Category"]
33         amt = exp["Amount"]
34
35         if cat in summary:
36             summary[cat] += amt
37         else:
38             summary[cat] = amt
39
40     print("\nCategory Summary")
41     for cat, total in summary.items():
42         print(cat, ":", total)
43
44 def budget_report():
45     total = sum(exp["Amount"] for exp in expenses)
46
47     print("\nBudget Report")
48     print("Total Spent:", total)
49     print("Budget:", budget)
50
51     if total > budget:
52         print("Budget Exceeded!")
53     elif total >= 0.8 * budget:
54         print("Warning: More than 80% budget used.")
55     else:
56         print("Budget Under Control.")
```

```
57
58 while True:
59     print("\n--- PERSONAL EXPENSE TRACKER ---")
60     print("1. Add Expense")
61     print("2. View Expenses")
62     print("3. Category Summary")
63     print("4. Budget Report")
64     print("5. Exit")
65
66     choice = input("Enter Choice: ")
67
68     if choice == "1":
69         add_expense()
70     elif choice == "2":
71         view_expenses()
72     elif choice == "3":
73         category_summary()
74     elif choice == "4":
75         budget_report()
76     elif choice == "5":
77         print("Thank You!")
78         break
79     else:
80         print("Invalid Choice")
```

Output:

```
OUTPUT  DEBUG CONSOLE  TERMINAL

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Enter Monthly Budget: 50000

--- PERSONAL EXPENSE TRACKER ---
1. Add Expense
2. View Expenses
3. Category Summary
4. Budget Report
5. Exit
Enter Choice: 1
Enter Description: Movie Ticket
Enter Category: Entertainment
Enter Amount: 300
Enter Date: 26-6-2026
Expense Added Successfully!

--- PERSONAL EXPENSE TRACKER ---
1. Add Expense
2. View Expenses
3. Category Summary
4. Budget Report
5. Exit
Enter Choice: 2

Expense List
1 {'Description': 'Movie Ticket', 'Category': 'Entertainment', 'Amount': 300.0, 'Date': '26-6-2026'}
```

--- PERSONAL EXPENSE TRACKER ---

1. Add Expense
2. View Expenses
3. Category Summary
4. Budget Report
5. Exit

Enter Choice: 3

Category Summary

Entertainment : 300.0

--- PERSONAL EXPENSE TRACKER ---

1. Add Expense
2. View Expenses
3. Category Summary
4. Budget Report
5. Exit

Enter Choice: 4

Budget Report

Total Spent: 300.0

Budget: 50000.0

Budget Under Control.

--- PERSONAL EXPENSE TRACKER ---

1. Add Expense
2. View Expenses
3. Category Summary
4. Budget Report
5. Exit

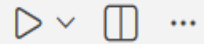
Enter Choice: 5

Thank You!

Project 4) Quiz and Examination System

Programming Code:

Quiz and Examination System ●



Quiz and Examination System > ...

```
1  questions = [  
2      ("Which data type is immutable in Python?",  
3      | "List", "Dictionary", "Tuple", " Set", "C"),  
4  
5      ("Which keyword is used to define a function?",  
6      | "func", "define", "def", "function", "C"),  
7  
8      ("Which symbol is used for comments in Python?",  
9      | "///", "#", "/*", "%", "B"),  
10  
11     ("Which data type stores key-value pairs?",  
12     | "List", "Tuple", "Dictionary", "Set", "C"),  
13  
14     ("Which loop is used for iteration in Python?",  
15     | "for", "repeat", "loop", "iterate", "A")  
16 ]  
17  
18 def conduct_quiz():  
19     name = input("Enter Student Name: ")  
20     roll = input("Enter Roll Number: ")  
21  
22     score = 0  
23     wrong_answers = []  
24  
25     print("\n===== QUIZ START =====")  
26
```

```
27     for i, q in enumerate(questions, start=1):
28         print(f"\nQ{i}. {q[0]}")
29         print("A.", q[1])
30         print("B.", q[2])
31         print("C.", q[3])
32         print("D.", q[4])
33
34         answer = input("Enter Answer (A/B/C/D): ").upper()
35
36         if answer == q[5]:
37             print("Correct Answer!")
38             score += 1
39         else:
40             print("Wrong Answer!")
41             wrong_answers.append({
42                 "Question": q[0],
43                 "Correct Answer": q[5]
44             })
45
46     total = len(questions)
47     percentage = (score / total) * 100
48
```



```
49     if percentage >= 80:
50         grade = "A"
51     elif percentage >= 60:
52         grade = "B"
53     elif percentage >= 40:
54         grade = "C"
55     else:
56         grade = "F"
57
58     print("\n==== RESULT REPORT =====")
59     print("Name :", name)
60     print("Roll No :", roll)
61     print("Score :", score, "/", total)
62     print("Percentage :", percentage, "%")
63     print("Grade :", grade)
64
65     if wrong_answers:
66         print("\nWrong Answers Review:")
67         for item in wrong_answers:
68             print("Question:", item["Question"])
69             print("Correct Answer:", item["Correct Answer"])
70
71     conduct_quiz()
```

Output:

OUTPUT

DEBUG CONSOLE

TERMINAL

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 Python + ✓   ...

Enter Student Name: Dhanashri

Enter Roll Number: 16

===== QUIZ START =====

Q1. Which data type is immutable in Python?

A. List

B. Dictionary

C. Tuple

D. Set

Enter Answer (A/B/C/D): C

Correct Answer!

Q2. Which keyword is used to define a function?

A. func

B. define

C. def

D. function

Enter Answer (A/B/C/D): C

Correct Answer!

Q3. Which symbol is used for comments in Python?

A. //

B. #

C. /*

D. %

Enter Answer (A/B/C/D): B

Correct Answer!

Q4. Which data type stores key-value pairs?

- A. List
- B. Tuple
- C. Dictionary
- D. Set

Enter Answer (A/B/C/D): C

Correct Answer!

Q5. Which loop is used for iteration in Python?

- A. for
- B. repeat
- C. loop
- D. iterate

Enter Answer (A/B/C/D): A

Correct Answer!

===== RESULT REPORT =====

Name : Dhanashri

Roll No : 16

Score : 5 / 5

Percentage : 100.0 %

Grade : A

Project 5) Inventory Management System

Programming Code:

```
Inventory Management System X
Inventory Management System > ...

1  inventory = {}
2  categories = set()
3  transactions = []
4
5  def add_product():
6      pid = input("Enter Product ID: ")
7
8      if pid in inventory:
9          print("Product ID already exists!")
10         return
11
12     name = input("Enter Product Name: ")
13     category = input("Enter Category: ")
14     price = float(input("Enter Unit Price: "))
15     quantity = int(input("Enter Quantity: "))
16     reorder = int(input("Enter Reorder Level: "))
17
18     inventory[pid] = {
19         "name": name,
20         "category": category,
21         "price": price,
22         "quantity": quantity,
23         "reorder": reorder
24     }
25
26     categories.add(category)
27     print("Product Added Successfully!")
28
29 def stock_in():
30     pid = input("Enter Product ID: ")
31
32     if pid in inventory:
33         qty = int(input("Enter Quantity to Add: "))
34         inventory[pid]["quantity"] += qty
35
36         transactions.append(f"IN - {pid} - {qty}")
37         print("Stock Added Successfully!")
38     else:
39         print("Product Not Found!")
```

```

40
41 def stock_out():
42     pid = input("Enter Product ID: ")
43
44     if pid in inventory:
45         qty = int(input("Enter Quantity to Remove: "))
46
47         if qty <= inventory[pid]["quantity"]:
48             inventory[pid]["quantity"] -= qty
49
50             transactions.append(f"OUT - {pid} - {qty}")
51             print("Stock Removed Successfully!")
52         else:
53             print("Insufficient Stock!")
54     else:
55         print("Product Not Found!")
56
57 def view_inventory():
58     if len(inventory) == 0:
59         print("No Products Available")
60         return
61
62     print("\n----- INVENTORY LIST -----")
63
64     for pid, data in inventory.items():
65         value = data["price"] * data["quantity"]
66
67         print("\nProduct ID :", pid)
68         print("Name      :", data["name"])
69         print("Category  :", data["category"])
70         print("Price     :", data["price"])
71         print("Quantity  :", data["quantity"])
72         print("Value     :", value)
73
74 def low_stock_alert():
75     print("\n----- LOW STOCK PRODUCTS -----")
76
77     found = False
78

```

```

78
79     for pid, data in inventory.items():
80         if data["quantity"] <= data["reorder"]:
81             print(pid, "-", data["name"],
82                   |(Qty:", data["quantity"], ")")
83             found = True
84
85     if not found:
86         print("No Low Stock Products")
87
88 def generate_report():
89     total_value = 0
90
91     for data in inventory.values():
92         total_value += data["price"] * data["quantity"]
93
94     print("\n===== INVENTORY REPORT =====")
95     print("Total Products :", len(inventory))
96     print("Categories :", len(categories))
97     print("Total Inventory Value :", total_value)
98
99 def view_transactions():
100     print("\n----- TRANSACTION LOG -----")
101
102     if len(transactions) == 0:
103         print("No Transactions")
104     else:
105         for t in transactions:
106             print(t)
107

```

```
108 while True:
109     print("\n===== INVENTORY MANAGEMENT SYSTEM =====")
110     print("1. Add Product")
111     print("2. Stock In")
112     print("3. Stock Out")
113     print("4. View Inventory")
114     print("5. Low Stock Alert")
115     print("6. Inventory Report")
116     print("7. Transaction Log")
117     print("8. Exit")
118
119     choice = input("Enter Choice: ")
120
121     if choice == "1":
122         | add_product()
123
124     elif choice == "2":
125         | stock_in()
126
127     elif choice == "3":
128         | stock_out()
129
130     elif choice == "4":
131         | view_inventory()
132
133     elif choice == "5":
134         | low_stock_alert()
135
136     elif choice == "6":
137         | generate_report()
138
139     elif choice == "7":
140         | view_transactions()
141
142     elif choice == "8":
143         | print("Program Ended")
144         | break
145
146     else:
147         | print("Invalid Choice!")
```

Output:

```
OUTPUT    DEBUG CONSOLE    TERMINAL

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===== INVENTORY MANAGEMENT
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Inventory Report
7. Transaction Log
8. Exit
Enter Choice: 1
Enter Product ID: 2
Enter Product Name: Laptop
Enter Category: Electronic
Enter Unit Price: 50000
Enter Quantity: 10
Enter Reorder Level: 5
Product Added Successfully

===== INVENTORY MANAGEMENT
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Inventory Report
7. Transaction Log
8. Exit
Enter Choice: 2
Enter Product ID: 2
Enter Quantity to Add: 10
Stock Added Successfully!
```


===== INVENTORY MANAGEMENT

1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Inventory Report
7. Transaction Log
8. Exit

Enter Choice: 3

Enter Product ID: 3

Product Not Found!

===== INVENTORY MANAGEMENT

1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Inventory Report
7. Transaction Log
8. Exit

Enter Choice: 4

----- INVENTORY LIST -----

Product ID	:	2
Name	:	Laptop
Category	:	Electronic
Price	:	50000.0
Quantity	:	20
Value	:	1000000.0

===== INVENTORY MANAGEMENT

1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Inventory Report
7. Transaction Log
8. Exit

Enter Choice: 5

----- LOW STOCK PRODUCTS -
No Low Stock Products

===== INVENTORY MANAGEMENT

1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Inventory Report
7. Transaction Log
8. Exit

Enter Choice: 6

===== INVENTORY REPORT =====

Total Products : 1

Categories : 1

Total Inventory Value : 10

===== INVENTORY MANAGEMENT

1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Inventory Report
7. Transaction Log
8. Exit

Enter Choice: 7

----- TRANSACTION LOG -----

IN - 2 - 10

===== INVENTORY MANAGEMENT

1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Inventory Report
7. Transaction Log
8. Exit

Enter Choice: 8

Program Ended